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Next Generation AI Researchers for Air Quality Excellence

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Data Management for Low-cost Sensors
Module 1: Data modeling and Databases

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Consortium

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INSTITUTE FOR ANTHROPOLOGICAL RESEARCH

theLisbonCouncil
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INAR
Institute for Atmospheric and Earth System Research

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FACULTY OF ELECTRICAL ENGINEERING, COMPUTER SCIENCE AND INFORMATION TECHNOLOGY OSIJEK

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Agenda

- Introduction to Databases
- Conceptual Data Model
- Relational Data Model
- Structured Query Language

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Short introduction to databases

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History of data storage and management

- Essential for humans to store and retrieve information/data to make informed decisions
  - Should be as efficient as possible
  - Categorization, no duplicates, relations, etc …

- Manual records of Sumerians are an early example of storing data (4000 BC using clay)

[archive.org]

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History of data storage and management

- From clay, we moved to papyrus, parchment to paper

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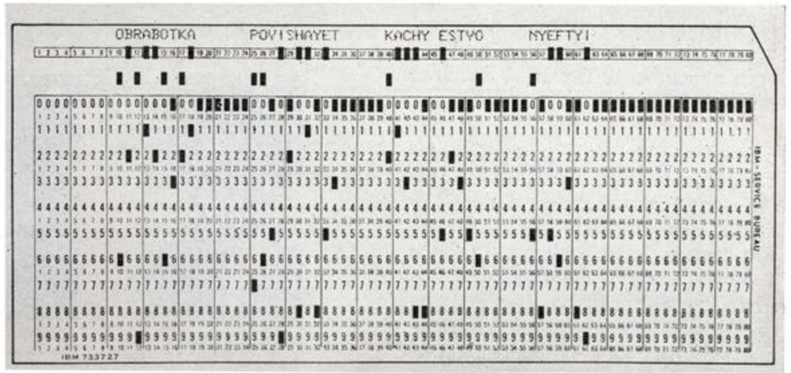
History of data storage and management

- From clay, we moved to papyrus, parchment to paper

- Data management in libraries → Dewey Decimal System
  - Developed around 1876 and still used today in 135 countries
  - Organizes books in libraries based on subjects
  - 3-digit numbers for main classes (e.g., 300 is social sciences)
  - Fractional decimals for detailed classes (e.g., 323 is civil and political rights)

- Punch cards (1900 – 1955)

- 1960 marks the start of computerized databases



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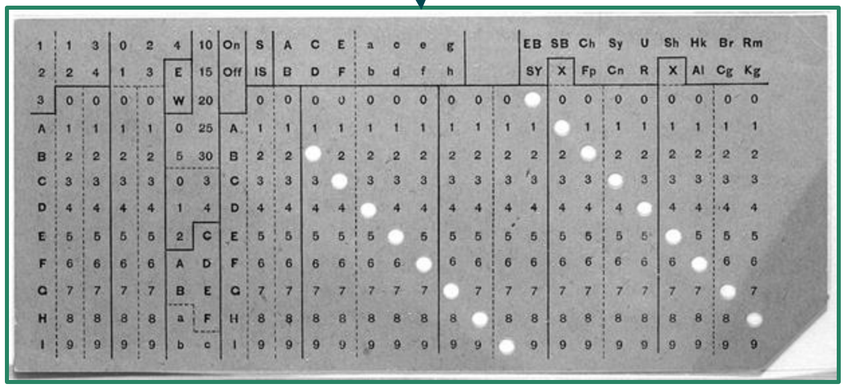
Computerized Databases

- **Flat files**
  - Flat → no connection with other files/records
  - E.g. Hollerit's punch cards database
  - Stored as sequential **files retrieval is slow**
  - Example: .csv file



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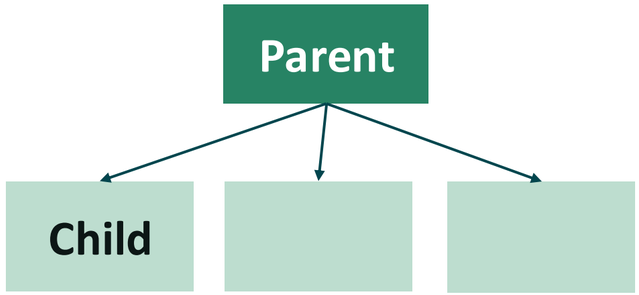
Computerized Databases

- **Flat files**
  - Flat → no connection with other files/records
  - Stored as sequential files → retrieval is slow
  - Example: .csv file

- **Hierarchical model**
  - Tree of records
  - E.g., IBM Information Management System
  - Each child can only have one parent



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Relational Databases (1970/80s - now)

School Table

ID	Name
S001	University of Technology
S002	University of Applied Science

Reference

Student Table

School ID	ID	Name	DOB
S001	UT-1000	Tommy	05/06/1995
S001	UT-1000	Better	16/04/1995
S002	UAS-1000	Linda	02/09/1995
S002	UAS-1000	Jonathan	22/06/1995

Tuple

Attribute

Relation (set of tuples with attributes, often the whole table)

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Relational Databases (1970/80s - now)

- Organize a body of data into simple tables of related information
- There are no pointers to maintain (reference)
- Tables connected only by having matching data field (keys)
- Easy to access, merge and change data
- Independence between data scheme and physical storage
- SQL (structured query language)

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Recommended DBMS: PostgreSQL

- Open-source and free relational database management system
- Easy to install, e.g., (https://www.postgresql.org/download/)
  - See also Tutorial pdf in NextAIRE learning platform
- Nice GUI (pgAdmin4)
- Good support and user community

[postgresql.org]

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What is a database?

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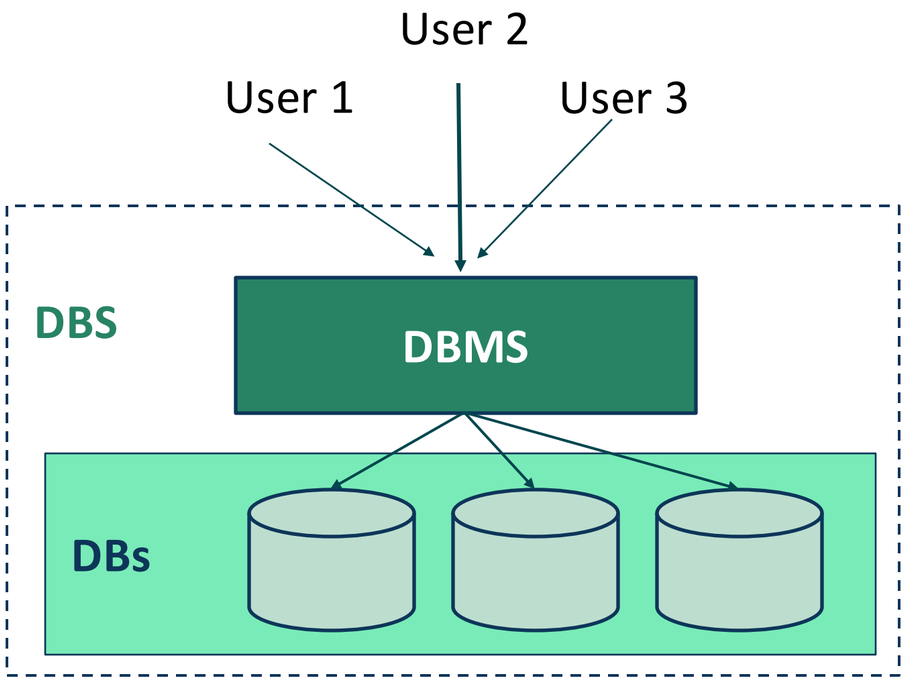
Definition of a database

- Database system (DBS): DBMS + DBs
- DBMS: Database Management System (Software to handle DBs)
- DBs: Database (data/metadata collection to describe something)
- Note: DB also a short for DBS/DBMS



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Database Design

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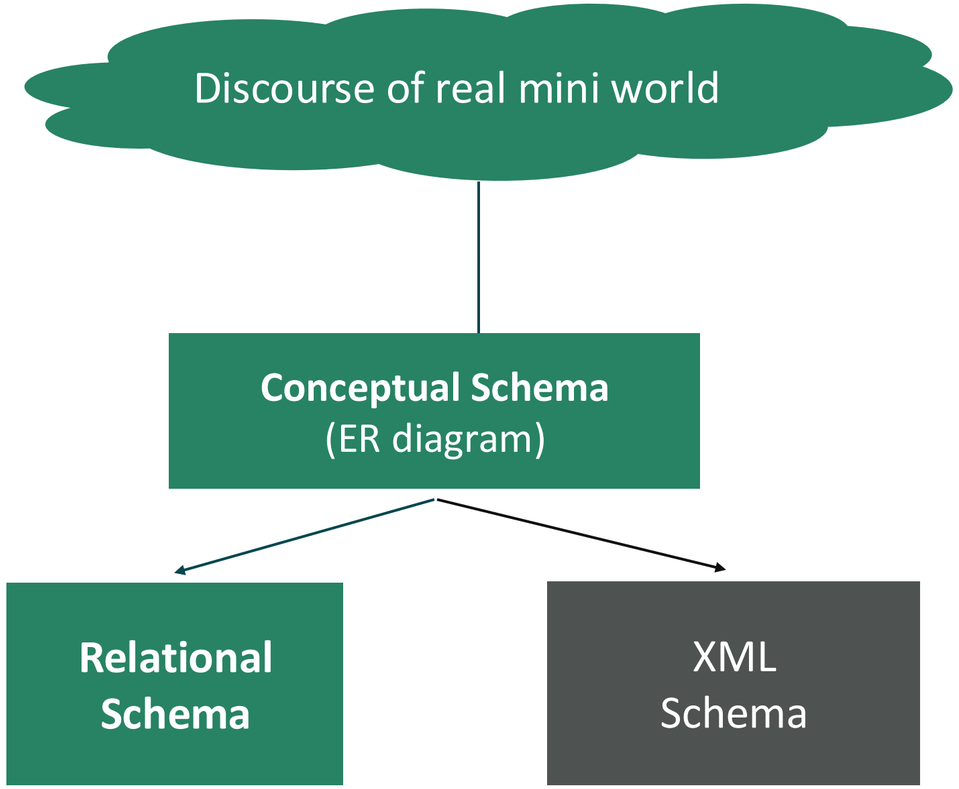
Data modeling

- Concept for describing data objects and their relationships
- Schema: Description (structure, semantics) of specific data collection



Manual Modeling

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Data models

- **Conceptual Data Models**
  - **Entity-Relationship Model (ERM)**, focus on data, ~1975
- **Logical Data Models**
  - **Relational data model**
  - Document (XML, JSON)
  - Time series

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Databases design cycles

- **Requirements engineering**
  - Collect and analyze data and application requirements
  - Specification documents
- **Conceptual Design**
  - Model data semantics and structure, independent of logical data model
  - ER model/ diagram
- **Logical Design**
  - Model data with implementation details of concrete data model (e.g., relational schema)
- **Physical Design**
  - Account for deployment environment and performance requirements

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Engineering Requirements

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Engineering Requirements

Collected Information

- Stations have a name and a location. They can host several sensors.
- Sensors have a SensorID, type, measured variable, and start date.
- Each sensor belongs to one station, while each station can contain many sensors.
- Campaigns have an ID, start date, and end date.
- Sensors can be used in monitoring campaigns.
- A sensor can be used in several campaigns, and a campaign can use several sensors.
- Some sensors can also validate campaigns.
- Each campaign is validated by one sensor, while one sensor can validate several campaigns.

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Conceptual Design

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Entity-relationship model/diagram

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ER Diagram - Chen notation

Entity (noun)

- Entities are objects of the real world
- Rectangular shape + capitalized

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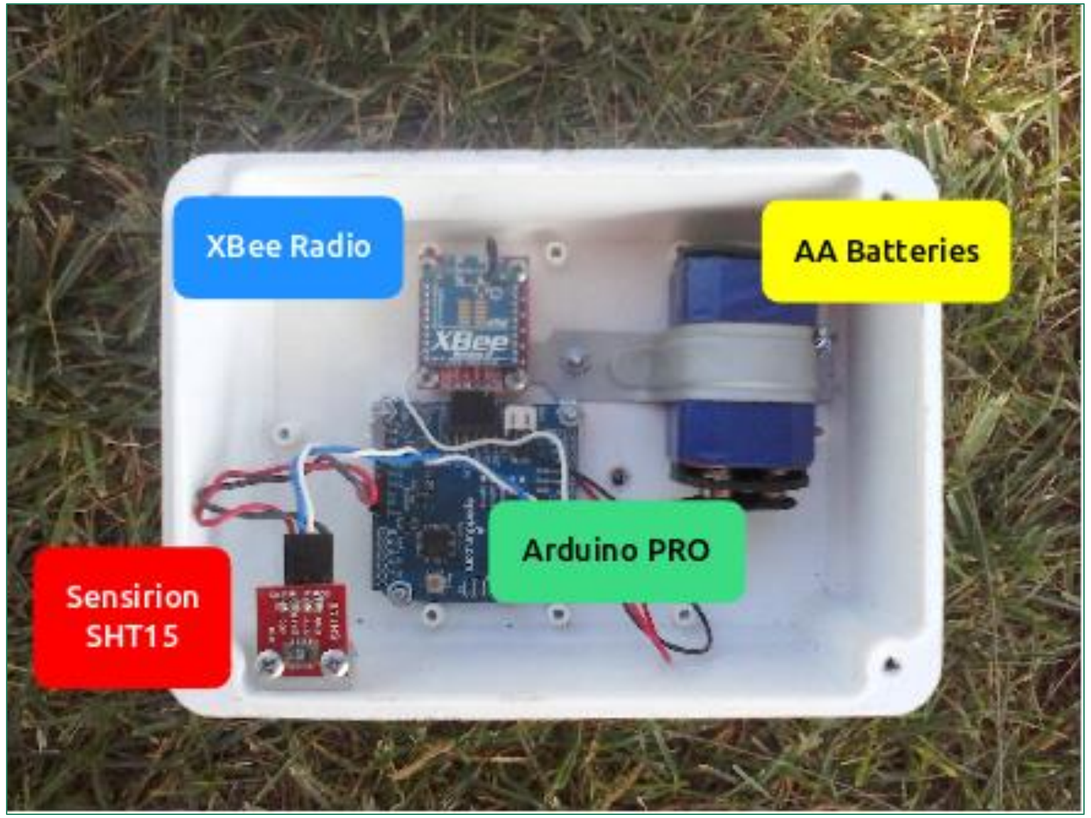
Entity Example: Sensor

Attributes	Value
sensor_id	SEN-001
name	Outdoor_weather
sensor_type	Humidity and temperature
sensor_model	Sensirion SHT15
channel_1	Relative humidity
channel_2	Temperature
humidity_range	0 – 100 %RH
humidity_accuracy	±2 %RH
temperature_range	-40 °C to 123 °C
temperature_accuracy	±0.3 °C



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ER Diagram - Chen notation

Entity (noun)

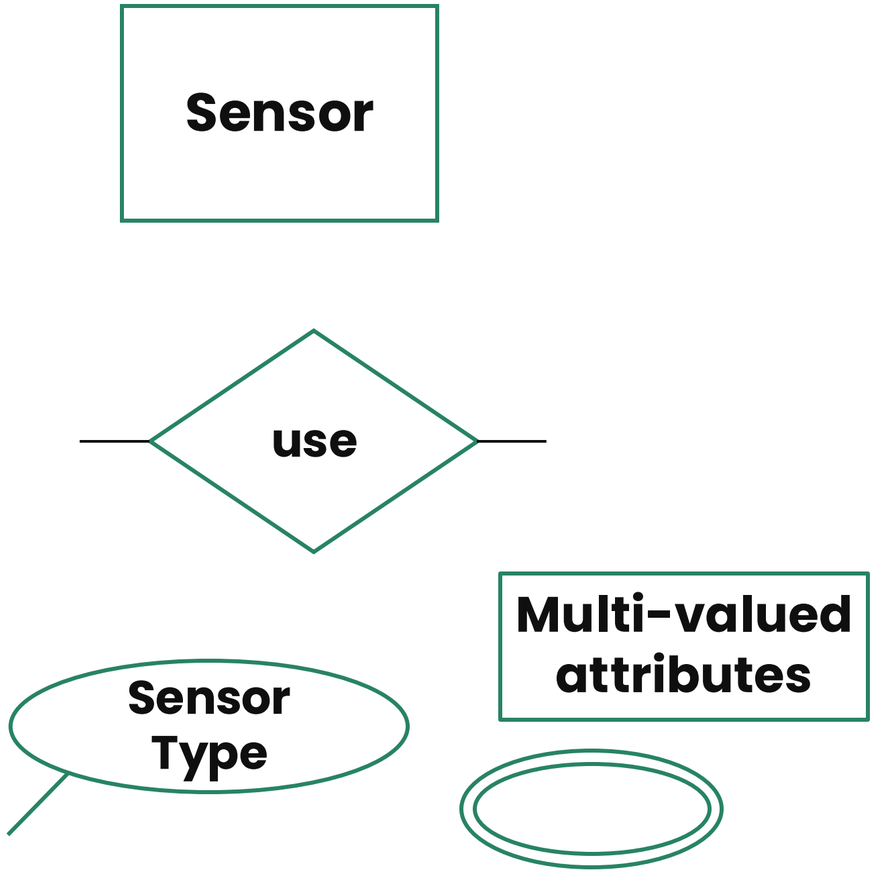
- Entities are objects of the real world
- An entity instance is a concrete instantiation of an entity
- Rectangular shape + capitalized

Relationship (verb)

- Relationships are concrete associations of entities
- Diamond shape + non-capitalized

Attribute

- Entities or relationships are characterized by attribute-value pairs
- **Round shape + capitalized**
- Multi-valued attributes (e.g., phone numbers)



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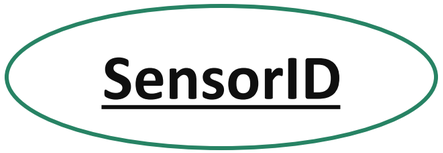
ER Diagram - Chen notation

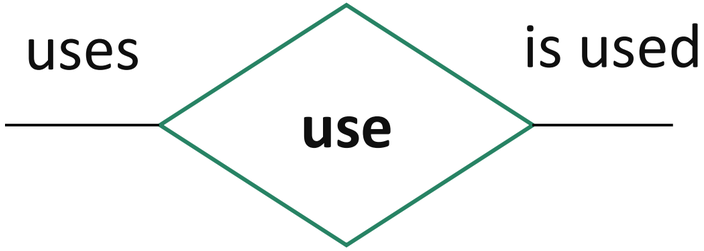
Keys

- Attributes that uniquely identify an entity
- Every entity type must have such a key
- Natural or surrogate (artificial) keys



Role

- Optional description of relationship types
- E.g., an sensor is used in a project & a project uses sensors



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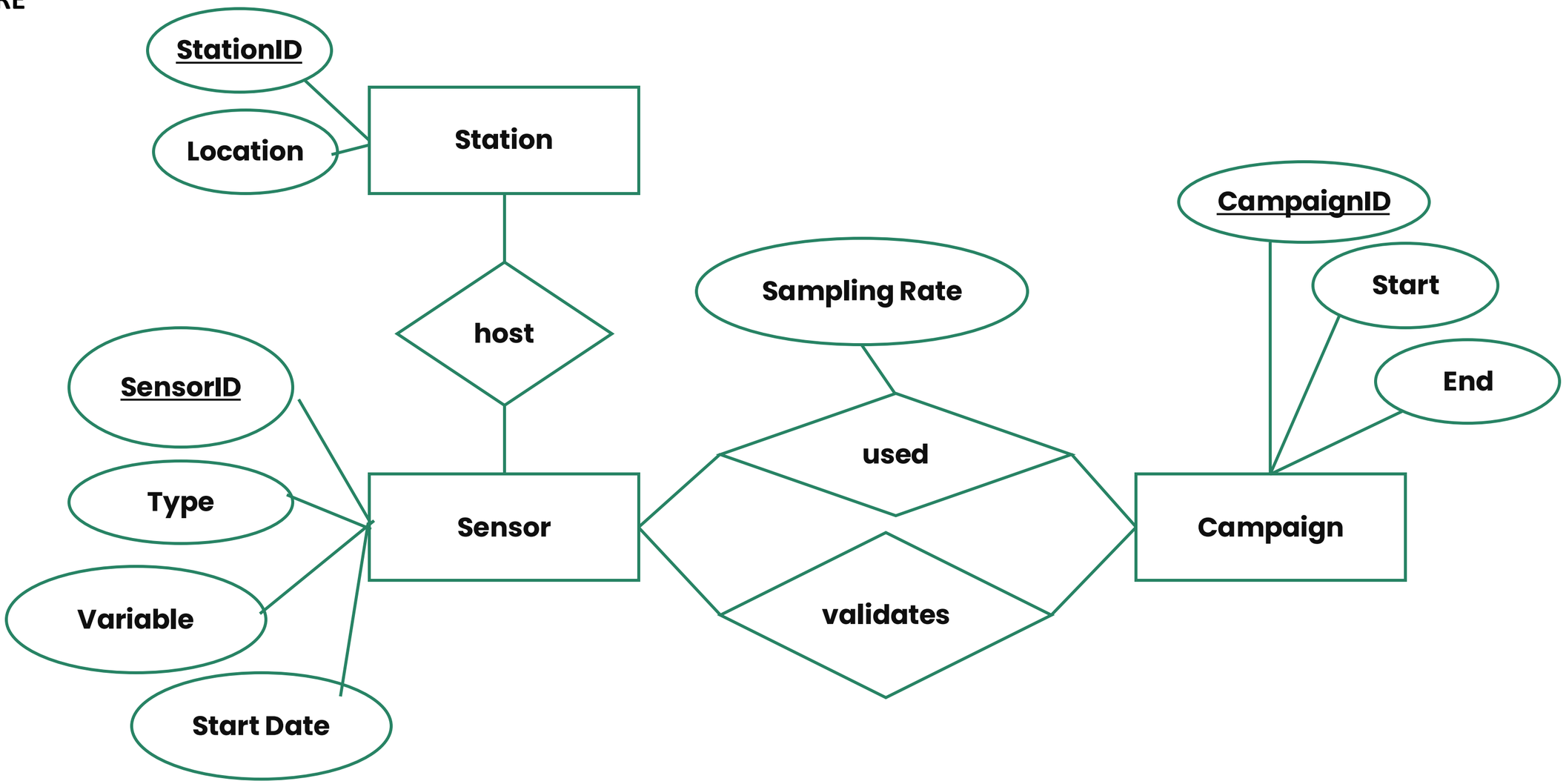
ER Diagram – A sensor database



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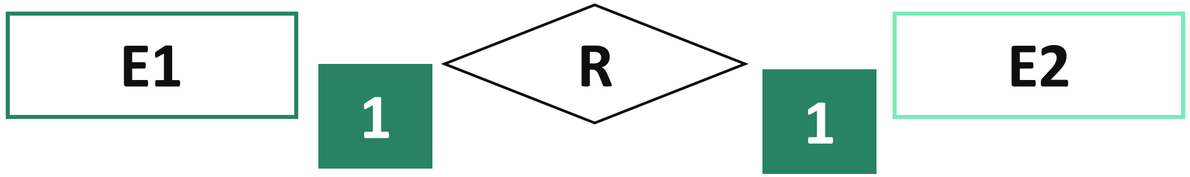
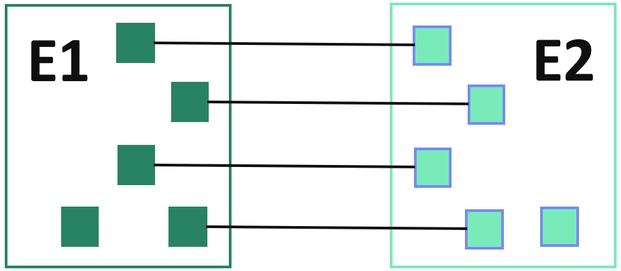
ER Diagram – Multiplicity/Cardinality





1:1 (one-to-one)

- Each Sensor relates to at most one Enclosure
- Each Enclosure relates to at most one Sensor
- 1 sensor can be installed in 1 enclosure and 1 enclosure can contain 1 sensor

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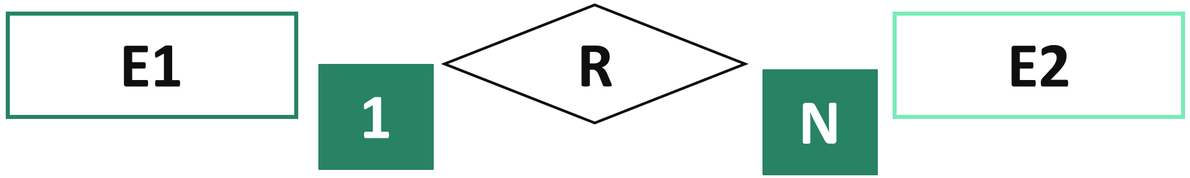
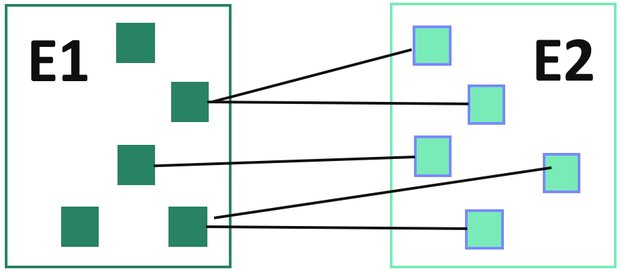
ER Diagram – Multiplicity/Cardinality





1:N (one-to-many)

- Each Station relates to many Sensors (0, 1, … N)
- Each Sensor relates to at most one Station
- E.g., 1 station can host N sensors but 1 sensor can only be installed at 1 station

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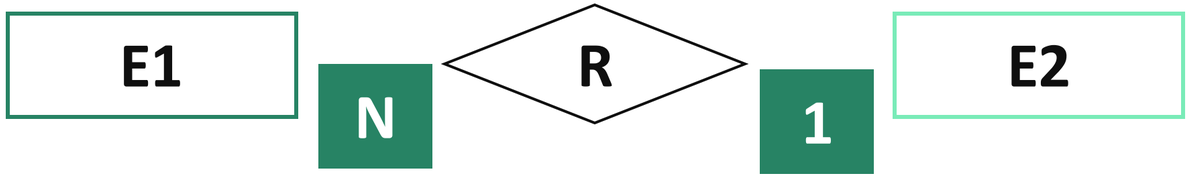
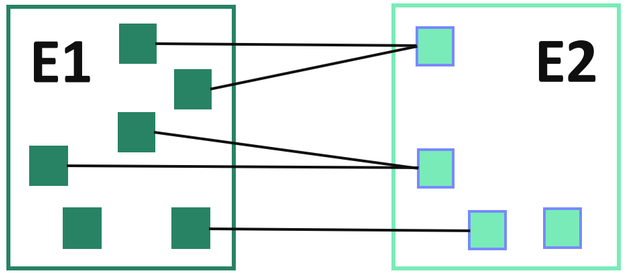
ER Diagram – Multiplicity/Cardinality





N:1 (many-to-one)

- Each Sensor relates to many Measurements (0, 1, … N)
- Each Measurement relates to at most one Sensor
- E.g., 1 sensor can record N measurements but 1 measurement can be recorded by just 1 sensor

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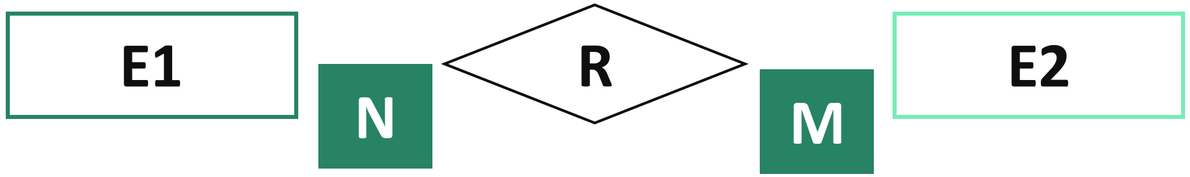
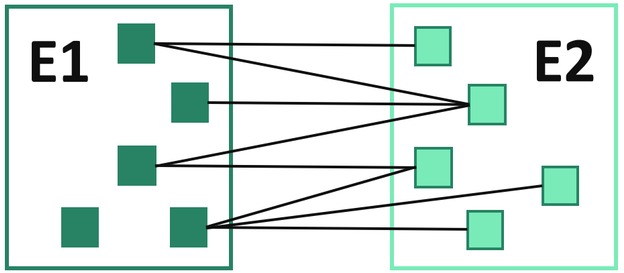
ER Diagram – Multiplicity/Cardinality



N:M (many-to-many)

- Each Sensor relates to many Campaigns (0, 1, … M)
- Each Campaign relates to many Sensors (0, 1, … N)
- E.g., 1 sensor can be used in M campaigns and 1 campaign can use N sensors



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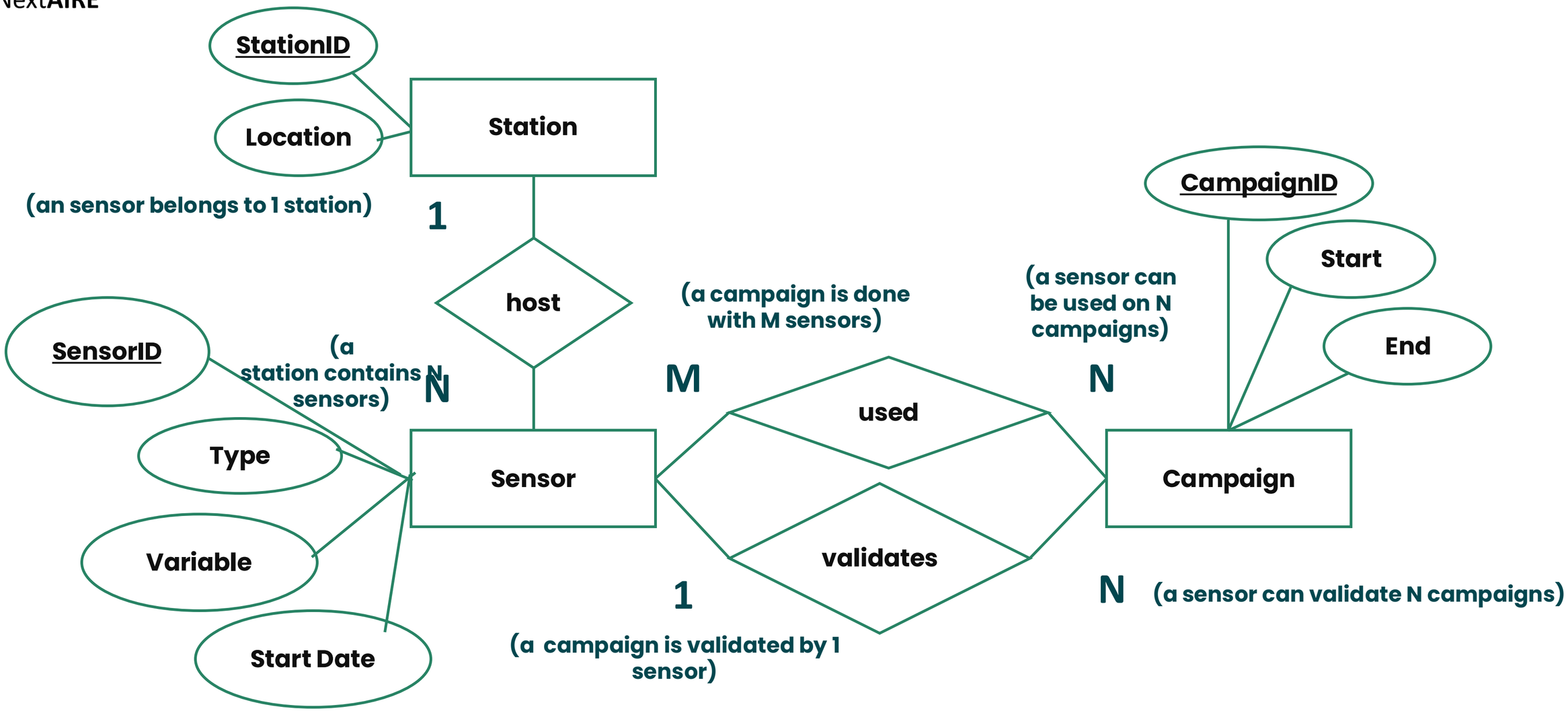
ER Diagram – Multiplicity/Cardinality



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Logical Design

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Relational Data Model

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Relational model

- Represent data as a **collection of tables**

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Relational model

- Represents a table (precisely a part of a table)
- Attributes represent the columns headers
- Tuples represent the rows
- Domain (value domain) represents a data type and set of values allowed for an attribute

My First Table

A1	A2	A3
3	7	T
1	2	T
3	4	F
1	7	T

Tuples

Attribute

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Data types in postgreSQL

Numeric type	Description
integer	4 bytes. Whole numbers from -2,147,483,648 to +2,147,483,647.
smallint	2 bytes. Smaller whole numbers.
bigint	8 bytes. Very large whole numbers.
serial	4 bytes. Auto-incrementing integer, useful for keys such as IDs.
real	4 bytes. Real values with about 6 decimal digits of precision.
double precision	8 bytes. Real values with about 15 decimal digits of precision.
numeric(precision, scale)	Exact decimal values. Example: 24.345 has precision 5 and scale 3.
numeric	Variable size when no precision or scale is provided.

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Data types in postgreSQL

Character type	Description
char(n)	Stores strings with a fixed length of n. Example: char(5) allows 'hello'. Longer strings cause an error. Shorter strings are padded with spaces.
char	Same as char(1). Stores one character.
varchar(n)	Stores strings with a maximum length of n. Longer strings cause an error. Shorter strings are not padded with spaces.
varchar	Stores strings of any length.
text	Stores strings of any length. In PostgreSQL, it is similar to varchar without a length limit.

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Data types in postgreSQL

Date/time type	Description
date	Stores a calendar date. Input is flexible, for example '1999-01-08' or '1999-Jan-08'. Common format: yyyy-MM-dd.
time	Stores a time of day, for example '04:05:06'. Common format: hh:mm:ss.
time with time zone	Stores a time of day with a time zone offset, for example '04:05:06+02'.
timestamp	Stores date and time together, for example '1999-01-08 04:05:06'. Common format: yyyy-MM-dd hh:mm:ss.
timestamp with timezone /timestamptz	Stores date and time with a time zone offset, for example '1999-01-08 04:05:06+02'.
to_timestamp()	Converts Unix timestamps into timestamp values. Unix timestamps count seconds since 1970-01-01.

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Data types in postgreSQL

Other types	**Description**
Boolean	For storing binary states true / false or 1 / 0 is possible
NULL	This is not a type but used to specify missing values for all types
XML / JSON	To store XML or JSON strings

Full list: https://www.postgresql.org/docs/current/datatype.html

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Domain

Domain (value domain) represents a **data type** and defines the **set of values allowed** for an attribute

Examples:

Age

- Can be represented with **Integer**
- 1,2,3,....,100

Name

- Can be represented as a **String** (array of characters)
- 'Jane Doe', 'John Doe'

Date of Birth

- Can be represented as a **String** or **Date**
- Can have **constraints** (**Date of Birth cannot be beyond the current date**)
- '21-08-1995', '02-07-2041'

A1 INT	A2 INT	A3 BOOL
3	7	T
1	2	T
3	4	F
1	7	T

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Primary Key

Primary Key: Minimal set of attributes X that uniquely identifies tuples in a relation R

#1 Unique: No other tuple with same value

#2 Defined: Not null

#3 Minimal: No attribute can be removed

sensor_id	type	variable	StationID
1	Sensirion SHT15	Temperature and humidity	GRZ01

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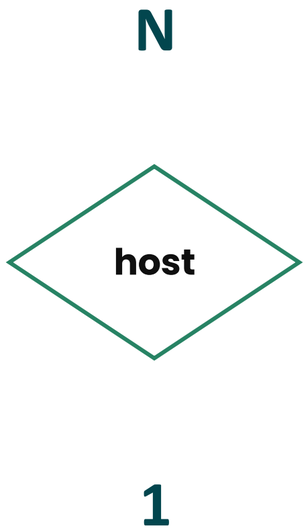
Foreign Key

Foreign Key: Reference to a primary key in another relation

sensor_id	type	variable	StationID
1	Sensirion SHT15	Temperature and humidity	GRZ01

StationID	Location
GRZ01	Austria



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Referential Integrity Constraints

StationID	Location
GRZ01	Austria
GRZ02	Austria
MDZ01	Argentina
EZE01	Argentina

sensor_id	type	variable	StationID
1	Sensirion SHT15	Temperature and humidity	GRZ01
2	DHT22	Temperature and Humidity	MDZ02
3	DHT22	Temperature and Humidity	GRZ01
4	Sensirion SHT15	Temperature and Humidity	GRZ02

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Referential Integrity Constraints

Foreign Keys:

- Reference to a **primary key** in **another relation**
- **Referential integrity:** FK **need to reference existing** tuples or **NULL**

Enforcing Referential Integrity

- **Error** (default)
- **Propagation** on request
  - E.g., for existential dependence
- **Set NULL** on request
  - E.g., for independent entities

```
DELETE FROM Stations WHERE Station ID=GRZ01
```

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Constraints

Domain/semantic constrains:

- Value constraints of individual attributes (e.g StationID lenght = 5 characters)
- **CHECK:** Value ranges or enumerated valid values (e.g humidity between 0 and 100)

- **UNIQUE Constraints**
  - Enforce uniqueness of non-primary key attribute
- **NOT NULL Constraints**
  - Enforce known / existing values
- **DEFAULT Constraints**
  - Sets a default value when created

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From ER diagram to Database schema

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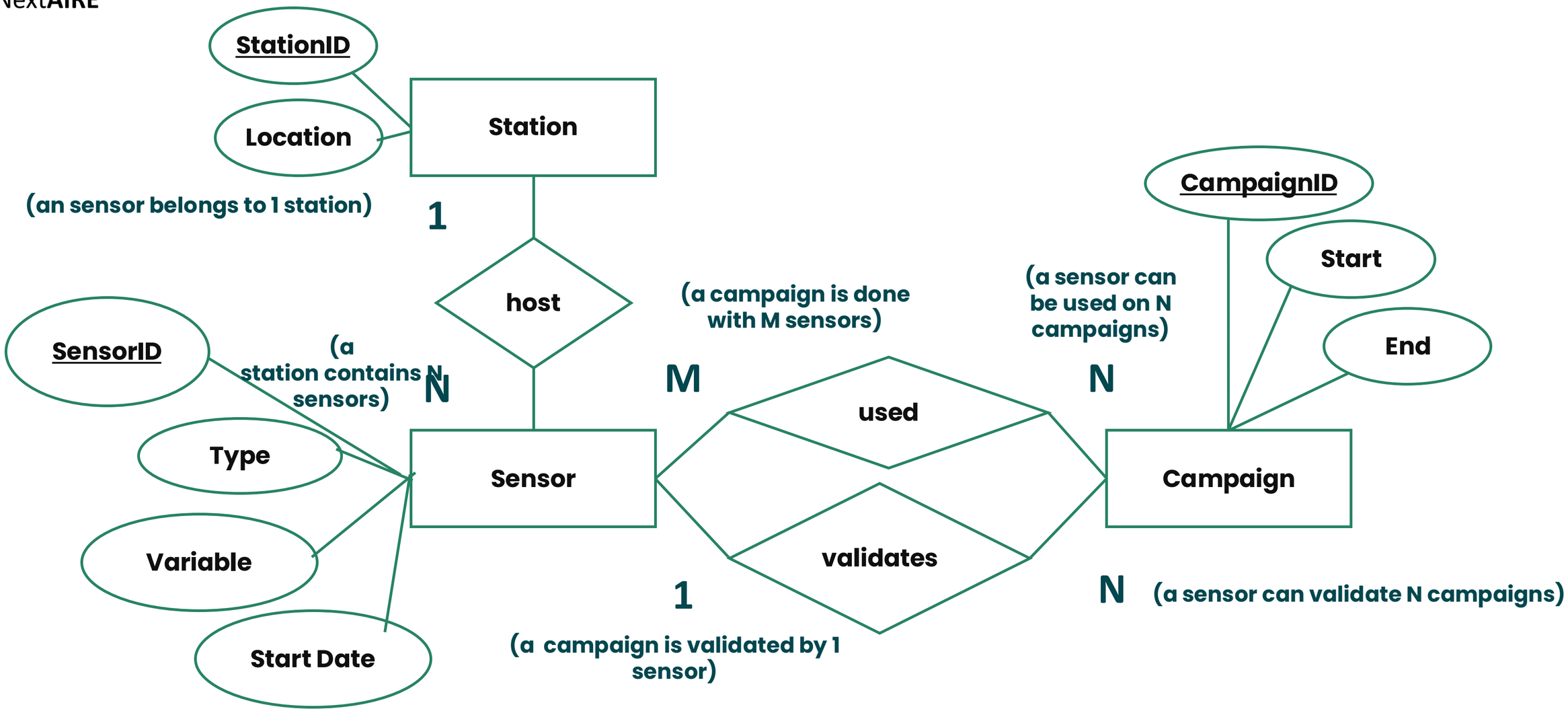
Sensor database



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Step 1: Mapping Entity Types

Each entity type directly maps to a relation:

Station	**Station**(StationID: INTEGER, Location: VARCHAR(256))
Sensor	**Sensor**(SensorID: INTEGER, Type: VARCHAR(128), Variable: VARCHAR(128), StartDate: DATE)
Campaign	**Campaign**(CampaignID: INTEGER, Start: DATE, End: DATE)

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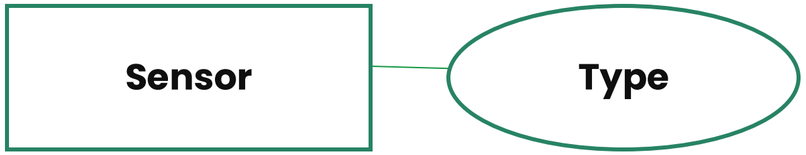
Step 2: Mapping Attributes

Atomic attributes:

- Direct mapping of ER attributes to attributes of relation
- Choice of data types and constraints



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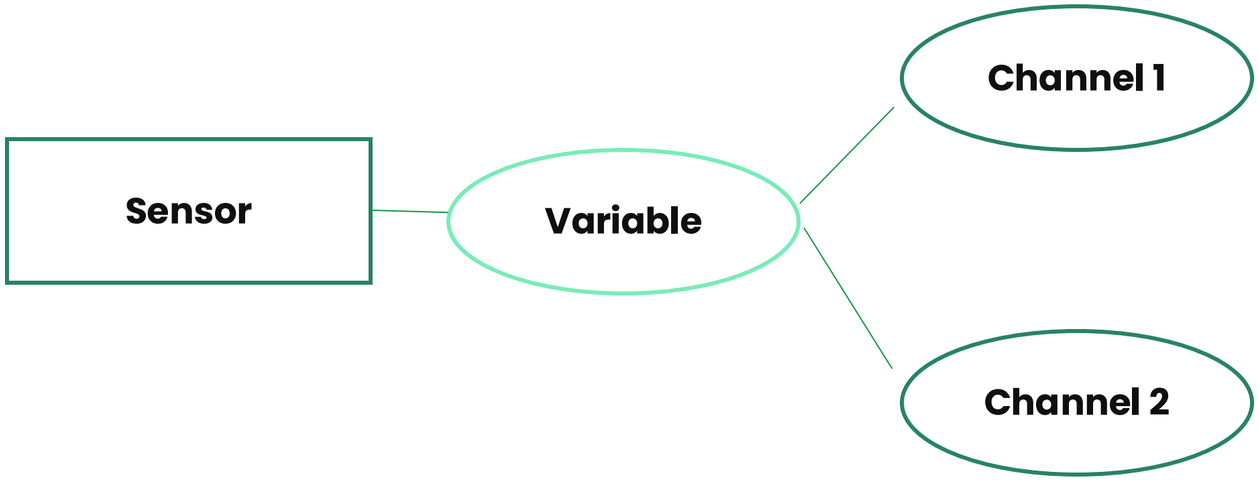
Step 2: Mapping Attributes

Composite attributes:

- Split into atomic attributes



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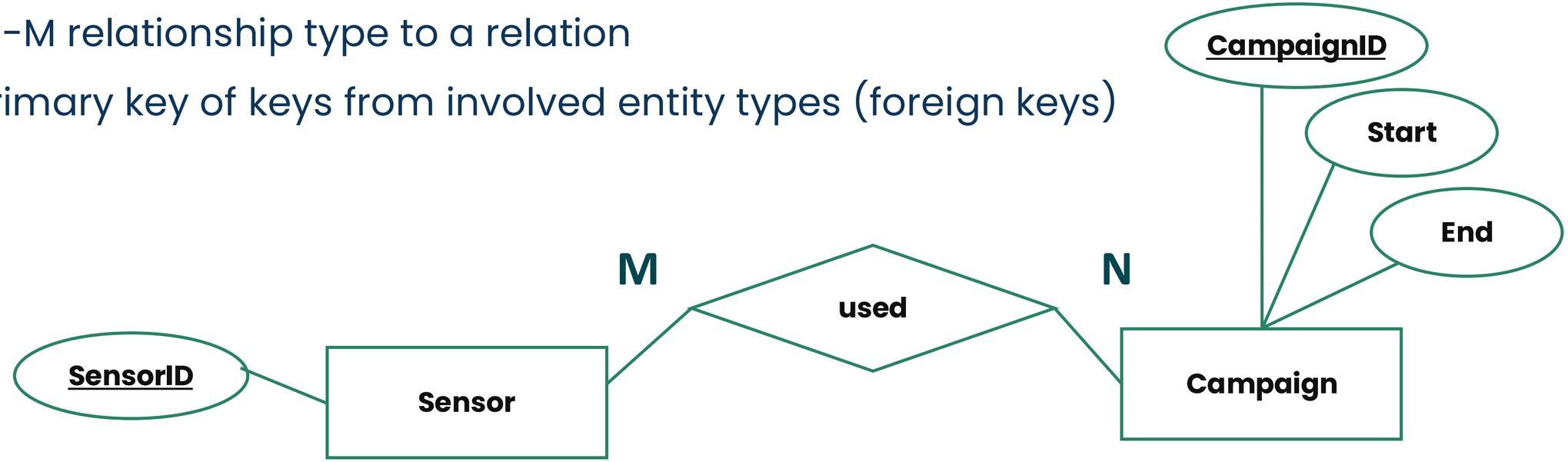
Step 2: Mapping Relationship Types

Generic Solution:

- Map every N-M relationship type to a relation
- Compose primary key of keys from involved entity types (foreign keys)



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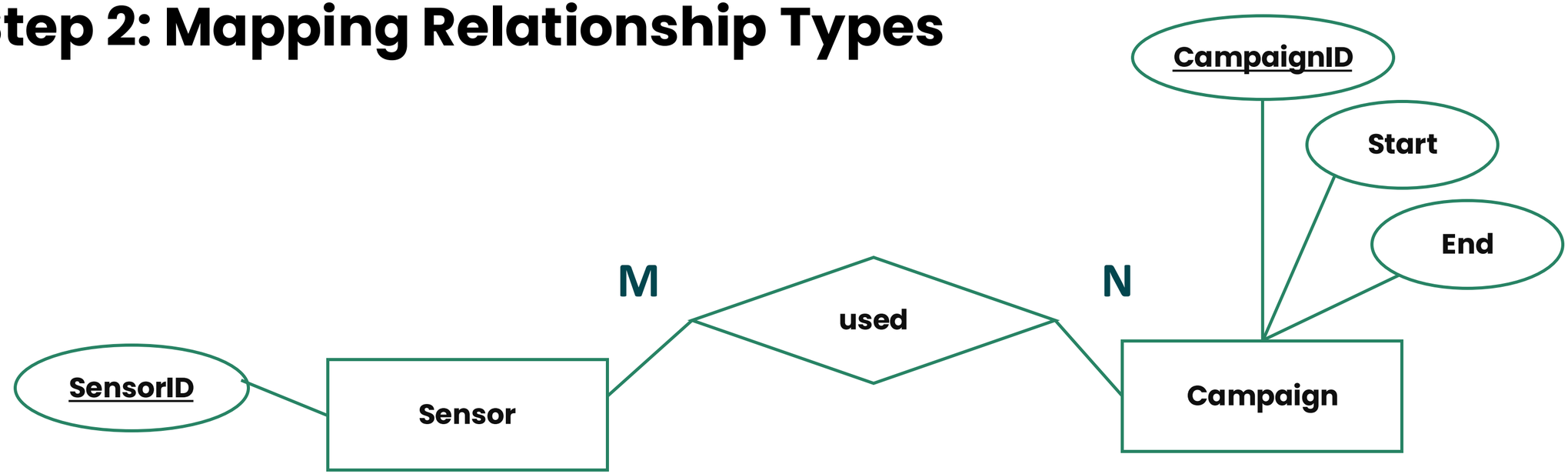
Step 2: Mapping Relationship Types



SensorID	CampaignID
1	12
2	12
3	01
4	07

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Structured Query Language (SQL)

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What is a SQL Query?

```
SELECT SensorID, Type, Variable, Location
FROM Sensor, Station
WHERE Sensor.StationID = Station.StationID
AND Location LIKE 'Austria';
```

SensorID	Type	Channel_1	Channel_2	StationID
1	Sensirion SHT15	Temperature	Humidity	GRZ01
2	DHT22	Temperature	Humidity	MDZ02
3	DHT22	Temperature	Humidity	GRZ01
4	Sensirion SHT15	Temperature	Humidity	GRZ02

StationID	Location
GRZ01	Austria
GRZ02	Austria
MDZ01	Argentina
EZE01	Argentina

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What is a SQL Query?

```
SELECT SensorID, Type, Channel_1, Channel_2, Location
FROM Sensor, Station
WHERE Sensor.StationID = Station.StationID
AND Location LIKE 'Austria';
```

SensorID	Type	Channel_1	Channel_2	Location
1	Sensirion SHT15	Temperature	Humidity	Austria
3	DHT22	Temperature	Humidity	Austria
4	Sensirion SHT15	Temperature	Humidity	Austria

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Overview SQL

Structured Query Language (SQL)

- **Data Definition Language (DDL)** → manipulate the database schema
- **Data Manipulation Language (DML)** → update and query database
- **Data Control Language (DCL)** → modify permissions

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Data Definition Language

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Create Tables

Structured Query Language (SQL)

- **Typed attributes**
- **Primary and foreign keys**
- **NOT NULL constraint**
- **→ Primary key has NOT NULL + UNIQUE constraints**

```
CREATE TABLE Sensor (
SensorID INTEGER PRIMARY KEY,
Type VARCHAR(128),
Channel_1 VARCHAR(128),
Channel_2 VARCHAR(128),
StationID VARCHAR(10),
FOREIGN KEY (StationID)
  REFERENCES Station(StationID) );
```

SensorID	Type	Channel_1	Channel_2	StationID
1	Sensirion SHT15	Temperature	Humidity	GRZ01
2	DHT22	Temperature	Humidity	MDZ02
3	DHT22	Temperature	Humidity	GRZ01
4	Sensirion SHT15	Temperature	Humidity	GRZ02

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Create Tables

StationID	Location
GRZ01	Austria
GRZ02	Austria
MDZ01	Argentina
EZE01	Argentina

```
CREATE TABLE Station (
StationID CHAR(5) PRIMARY KEY,
Location VARCHAR(128) NOT NULL
);
```

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Alter Tables

ADD/DROP columns

ALTER data type, defaults, constraints, etc

DROP TABLE → at the beginning of CREATE TABLE scripts

```
DROP TABLE IF EXIST Sensors, Stations, Campaign;
```

```
ALTER TABLE Sensor ADD Start Date DATE;
ALTER TABLE Sensor DROP Channel_2;

ALTER TABLE Campaign ADD CONSTRAINT
PRIMARY KEY (CampaignID);
```

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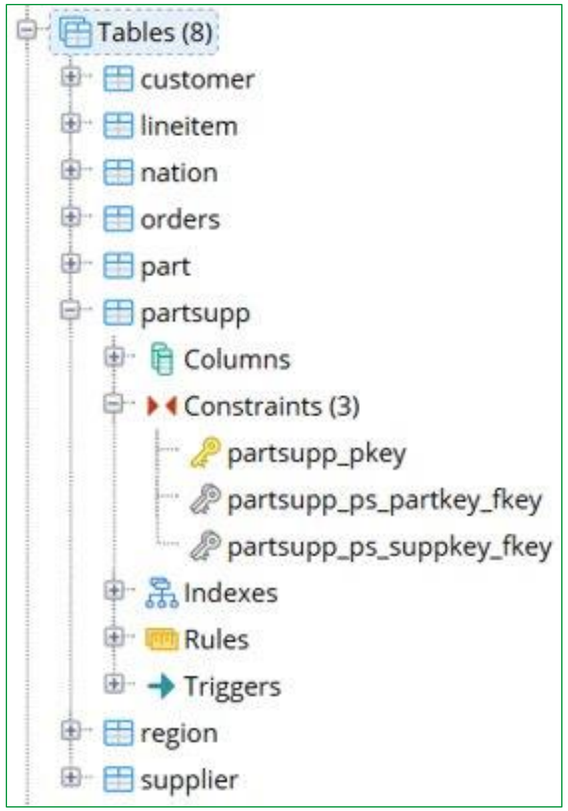
Database Catalog

Catalog Overview

- **Metadata** of all database objects (e.g., tables, constraints) → **read-only**
- **Accessible through SQL**
- Organized by schemas

```
SELECT * FROM information_schema.tables;
```



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Data Manipulation Language

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Insert Data

INSERT tuple

- **Single tuple with implicit or explicit attribute assignment**
- **No need to specify attributes when you add values for all attributes**

```
INSERT INTO Sensor (SensorID, Type, Channel_1, Channel_2, StationID)
VALUES (1, 'Sensirion SHT15', 'Temperature', 'Humidity', 'GRZ01'); --Explicit
```

```
INSERT INTO Sensor VALUES (1, 'Sensirion SHT15', 'Temperature', 'Humidity', 'GRZ01'); --Implicit
```

```
INSERT INTO Sensor
VALUES
(1, 'Sensirion SHT15', 'Temperature', 'Humidity', 'GRZ01'), --Multiple tuples
(2, 'DHT22', 'Temperature', 'Humidity', 'MDZ02'),
(3, 'DHT22', 'Temperature', 'Humidity', 'GRZ01');
```

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Insert Data

Insert from CSV file

Read and insert from a file-name using **COPY**

- Attributes in the CSV file need to match the DB table
- file-name is the path e.g., 'C:\Users\UserName\FolderName\file.csv'
- Arguments: **DELIMITER, CSV, HEADER (file with a header)**

```
COPY table-name FROM file-name CSV HEADER DELIMITER ',';
VALUES
/* this is a comment to document what is happening across multiple lines – make sure that user
“everyone” from Windows has read access to the files */
```

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Update and Delete

UPDATE Tuple/Table

- **Update of attributes**
- **Without WHERE all tuples are updated!**

DELETE Tuple/Table

- **Delete of tuples**
- **Without WHERE all tuples are deleted!**

```
UPDATE Sensor
SET Type = 'DHT22';
WHERE SensorID = 1;

DELETE FROM Sensor
WHERE SensorID = 1;
```

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Basic Queries

```
SELECT [DISTINCT] <column_list>
  FROM [<table_list> |
    <table1> [RIGHT | LEFT | FULL] JOIN
    <table2> ON <condition>]
  [WHERE <predicate>]
  [GROUP BY <column_list>]
    [HAVING <grouping predicate>]
  [ORDER BY <column_list> [ASC | DESC]];
```

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Basic Queries: SELECT

Distinct and All

- Distinct = No duplicates
- All = Duplicates included

```
SELECT DISTINCT ID, Type
  FROM Sensor;
```

Aliases

- Give table or column another name
- Can then be used in the query (e.g. S.Channel_1)

```
SELECT DISTINCT Type AS T,
  FROM Sensor AS S;
```

Select * → all columns

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Basic Queries: FROM – INNER JOIN

Specifies the tables

- Possible to combine rows from tables with related columns using JOINS
- Used for foreign key relationships

```
SELECT DISTINCT ID, Type
FROM Sensor;
```

(Inner) Join

- Get Sensor ID, Sensor Type and Station Location of all sensors

```
SELECT S.SensorID, S.Type, T.Location
FROM Sensor AS S;
INNER JOIN Station AS T
ON S.StationID = T.StationID:
```

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Basic Queries: FROM – INNER JOIN

```
SELECT S.SensorID, S.Type, T.Location
FROM Sensor AS S;
INNER JOIN Station AS T
ON S.StationID = T.StationID:
```

SensorID	Type	Channel_1	Channel_2	StationID
1	Sensirion SHT15	Temperature	Humidity	GRZ01
2	DHT22	Temperature	Humidity	MDZ02
3	DHT22	Temperature	Humidity	GRZ01
4	Sensirion SHT15	Temperature	Humidity	GRZ02

StationID	Location
GRZ01	Austria
GRZ02	Austria
MDZ01	Argentina
EZE01	Argentina

SensorID	Type	Location
1	Sensirion SHT15	Austria
3	DHT22	Austria
4	Sensirion SHT15	Austria

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Basic Queries: FROM – INNER JOIN

```
SELECT S.SensorID, S.Type, T.Location
FROM Sensor AS S;
INNER JOIN Station AS T
ON S.StationID = T.StationID:
```

SensorID	Type	Channel_1	Channel_2	StationID
1	Sensirion SHT15	Temperature	Humidity	GRZ01
2	**DHT22**	**Temperature**	**Humidity**	**MDZ02**
3	DHT22	Temperature	Humidity	GRZ01
4	Sensirion SHT15	Temperature	Humidity	GRZ02
5	**Sensirion SHT71**	**Temperature**	**Humidity**	**NULL**

StationID	Location
GRZ01	Austria
GRZ02	Austria
MDZ01	Argentina
EZE01	Argentina

Sensor 2 → FK not found in Station

NULLS → Omitted in output

SensorID	Type	Location
1	Sensirion SHT15	Austria
3	DHT22	Austria
4	Sensirion SHT15	Austria

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Basic Queries: FROM – OUTER JOIN

```
SELECT S.SensorID, S.Type, T.Location
FROM Sensor AS S;
LEFT OUTER JOIN Station AS T
ON S.StationID = T.StationID:
```

SensorID	Type	Channel_1	Channel_2	StationID
1	Sensirion SHT15	Temperature	Humidity	GRZ01
2	**DHT22**	**Temperature**	**Humidity**	**MDZ02**
3	DHT22	Temperature	Humidity	GRZ01
4	Sensirion SHT15	Temperature	Humidity	GRZ02
5	**Sensirion SHT71**	**Temperature**	**Humidity**	**NULL**

StationID	Location
GRZ01	Austria
GRZ02	Austria
MDZ01	Argentina
EZE01	Argentina

All non-matching fields **(in LEFT)** are filled with NULL **(from RIGHT)**

SensorID	Type	Location
1	Sensirion SHT15	Austria
2	**DHT22**	**NULL**
3	DHT22	Austria
4	Sensirion SHT15	Austria
5	**Sensirion SHT71**	**NULL**

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Basic Queries: WHERE

To add conditions to the query

- Combine conditions with AND / OR
- WHERE NOT to negate conditions
- Single quotes for text fields
- Wild cards for LIKE (% placeholder for an char)
- **Full list of operators:**

  https://www.w3schools.com/sql/sql_where.asp

```
SELECT SensorID, Type, Channel_1, Channel_2,
StationID
FROM Sensor
    WHERE StationID = 'GRZ01' OR StationID = 'GRZ02';
```

```
SELECT SensorID, Type, Channel_1, Channel_2,
StationID
FROM Sensor
    WHERE Type LIKE 'Sensirion%';
```

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Grouping and Aggregation

GROUP BY

- Allows to summarize data by categories
- Summarize → aggregates (MAX, MIN, AVG, COUNT)

```
SELECT StationID, COUNT(*) AS NumberOfSensors
FROM Sensor
GROUP BY StationID;
```

How many sensors are installed at each station?

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Grouping and Aggregation

HAVING

- Filter groups after calculation of aggregates

```
SELECT StationID, COUNT(*) AS NumberOfSensors
FROM Sensor
GROUP BY StationID
HAVING COUNT (*) > 1;
```

Which stations have more than one sensor?

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Sorting

Sorting

- Convert output into a sorted list of tuples
- **DESC** → from highest value to lowest
- **ASC** → from lowest value to highest
- Multiple fields possible
- Order by first and in case of ties by the second, etc.
- Evaluated last in a query

```
SELECT SensorID, Type, Channel_1, Channel_2,
StationID
FROM Sensor
ORDER BY StationID ASC, Type ASC;
```

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Data Control Language

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Data Control Language

Set access permissions on tables

- **GRANT** query/modification rights on database objects for specific users, roles
- **REVOKE** access rights from users, roles
- Same syntax for **UPDATE**, **INSERT**, etc

```
GRANT SELECT
ON TABLE Sensor
TO liacono;
```

```
REVOKE SELECT
ON TABLE Sensor
TO liacono;
```

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Thank You!

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